

Issa GUEYE

PhD Candidate in Applied Mathematics - Scientific Computing / HPC Engineer - Data Scientist



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📍 Dakar / Rufisque, Senegal 📞 +221 77 509 59 82

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@ issa.gueye@cineri.sn

About Me

Curious researcher.
Solid background in
mathematics, computer
science and Big Data.
Passionate about HPC and
scientific computing.
Curious explorer
Critical thinker
Creative innovator
Effective collaborator
Clear communicator
Problem solver
Adaptable and resilient
Rigorous and ethical
Optimistic yet realistic

Specialties

High Performance Computing
(HPC)
Scientific Computing &
Modeling
Parallel Programming
(C/C++, Fortran, MPI/OpenMP)
GPU Computing (CUDA,
Kokkos)
Machine & Deep Learning
Bayesian Inference
Generative Models (VAE,
Transformers)
Statistical Genomics
Data Analysis & Visualization
Statistical Modeling

Skills

- HPC & Parallel Programming
 - Scientific computing & modeling
- Python, R, C, C++, Fortran
- GPU Computing (CUDA, Kokkos)
 - SIMD Vectorization (AVX2/AVX-512)
 - Profiling & memory optimization
- HPC and Cloud Computing
- Distributed infrastructures
 - Algorithm optimization
- Large-scale data analysis
 - Statistical & predictive modeling
- Deep generative models
 - Hierarchical Bayesian inference
- Scientific publication writing

Tools

Python / PyTorch/JAX
R / RStudio / SQL
C/C++ / Fortran
CUDA / Kokkos / OpenMP

CAREER OBJECTIVE

PhD Candidate in Applied Mathematics (UGB-IPD-CINERI) and Junior Scientific Computing Engineer at CINERI, holder of a Master's degree in Mathematical Sciences, Big Data track (AIMS Senegal, First Class Honours) and a Master's degree in Mathematics and Applications, Data Science and Applications track (Université Gaston Berger, Upper Second Class Honours). My doctoral research focuses on developing genotype imputation methods for African populations, combining hierarchical Bayesian modeling, deep generative models (VAE, Transformers, diffusion models) and high-performance computing. Proven expertise in CPU/GPU optimization, SIMD vectorization, OpenMP/MPI parallelization and programming on heterogeneous architectures (CUDA, Kokkos). My goal is to contribute to the development of innovative solutions that strengthen African scientific research and national digital infrastructures.

PROFESSIONAL EXPERIENCE

Apr 2026 – Present

Junior Scientific Computing Engineer

CINERI · Diamniadio, Senegal 📍

- Supporting researchers in parallelizing and optimizing scientific codes (MPI, OpenMP, CUDA)
- Administration and operation of the national HPC cluster (Slurm, modules, pipelines)
- CPU optimization: SIMD vectorization (AVX2/AVX-512), memory hierarchy management, performance profiling
- Porting code to heterogeneous GPU architectures (CUDA, Kokkos)
- Testing and evaluation of HPC software and hardware solutions
- Contributing to high-performance computing projects and writing technical documentation



Oct 2025 – Apr 2026

Scientific Computing Engineer Intern

CINERI · Diamniadio, Senegal 📍

- Parallelization and optimization of scientific codes on HPC cluster
- Programming and optimization on heterogeneous architectures (CPU/GPU)
- Setting up Nextflow/Singularity pipelines for reproducibility



Aug 2024 – Feb 2025

Data Scientist / Statistical Analyst

INSTITUT PASTEUR DE DAKAR · Dakar, Senegal 📍

- Analysis of large-scale genomic and epidemiological datasets (Dielmo-Ndiop project)
- Development of statistical and machine learning models for infectious disease modeling
- Optimization of predictive algorithms and bioinformatics pipelines



Bash / Slurm / GitHub
Nextflow / Singularity
PLINK2 / bcftools
IMPUTE5 / Beagle
perf / Valgrind / gprof
LaTeX / Linux

Social Networks

 Issa Gueye

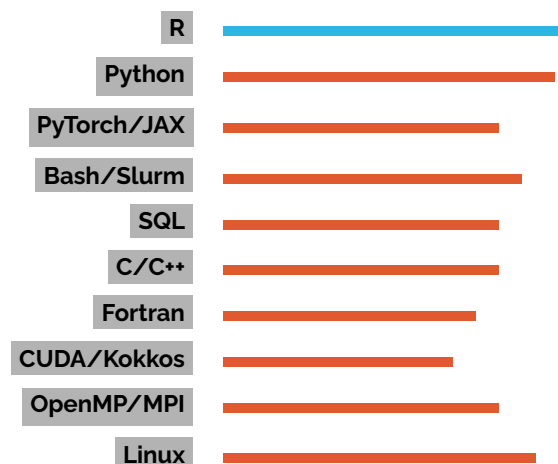
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 @IssaIsaacNewton

ONGOING DOCTORAL PROJECT

2025–2028 PhD in Applied Mathematics — Université Gaston Berger (LERSTAD)
Development of a genotype imputation method for genome-wide association studies in African populations: a hierarchical Bayesian approach, deep generative models, and high-performance computing
Advisor: Pr. Abdou Ka Diongue (UGB)
Co-advisor: Dr. Cheikh Loucoubar (IPD)
Supervisor: Dr. Aboubacry Gaye (IPD)
Labs: LERSTAD (UGB) - PESD/Institut Pasteur de Dakar - CINERI

PROGRAMMING



LANGUAGES

French	C2	Native language
Wolof	C2	Native language
English	C1	
Spanish	A2	

CERTIFICATIONS

2026 Programming and Optimization on Heterogeneous Architectures
CPU (C++/Fortran/Python, SIMD, OpenMP) and GPU (CUDA, Kokkos, JAX, PyTorch)
LAPP/CNRS, CC-FR, EuroHPC – CINERI Site, Senegal

2025 IBM Data Science with Python

2025 Google Data Scientist Certificate

2025 DataCamp – Data Science Career Track

2025 AWS Cloud Manager

2023 Personal Development & Digital Entrepreneurship (AIMS)

2021 Leadership & Entrepreneurship (Saint-Louis)

2020 Personal Development (Hub-Nord, Saint-Louis)

CONFERENCES & PRESENTATIONS

2026 SDSAC Workshop — AIMS Senegal (submitted)

2025 Machine Learning Summer School (MLSS)

2025 International Conference on Probability and Statistics

2023 "Option Pricing with Black-Scholes Extensions", AIMS Senegal

2022 "Introduction to LaTeX", UGB Mathematics Club

2021 "First Steps with Python", UGB Mathematics Club

EDUCATION

2025 Master's, Big Data
AIMS · Senegal 
Big Data & Mathematical Sciences

2023 Master's, M.A.
UGB · Senegal 
Data Science & Finance

2021 Bachelor's, M.A.S.S.
UGB · Senegal 
Applied Mathematics

2016 Baccalaureate, S1
LYCÉE ABDOULAYE SADJI · Senegal 

DEGREES

2016 Scientific Baccalaureate (Series S1)

2017–2021 B.Sc. M.A.S.S. — Lower Second Class Honours

2021–2023 Master's in Data (UGB) — Upper Second Class Honours

2023–2025 Master's in Big Data (AIMS) — First Class Honours

2025–2028 PhD in Applied Mathematics (UGB) — in progress